

Customer Engagement & Service Analytics Platform

Capstone Project Documentation

Table of Contents: Modules – 1,2,3,4,5,6,7,8,9

1. Executive Summary
2. Business Context & Problem Statement
3. Project Objectives
4. System Architecture
5. Technology Stack
6. Database Design & Schema
7. Backend Design & API Implementation
8. Frontend Architecture & UI Design
9. Ticket Workflow & Business Rules
10. Validation & Error Handling Strategy
11. Security & Authentication
12. Analytics & Dashboard Logic
13. Deployment Strategy
14. Testing & Quality Assurance
15. Challenges Faced & Resolutions
16. Future Enhancements
17. Power BI Dashboard Implementation

17.1 Data connection

17.2 KPI Measures(DAX)

17.3 Dashboard values

17.4 Interactive filters (Slicers)

17.5 Layout and Design

18. Insights supported by Dashboard

19. Conclusion

1. Executive Summary

The Customer Engagement & Service Analytics Platform is a modular, scalable, and enterprise-ready solution developed to modernize service ticket management for ConnectPlus Support Services. The system centralizes customer and ticket operations, enforces strict workflow control, ensures database integrity, and provides real-time analytics dashboards.

2. Business Context & Problem Statement

ConnectPlus Support Services handles high volumes of customer inquiries. Operational inefficiencies existed due to duplicate tickets, inconsistent status updates, manual tracking, and lack of reporting capabilities.

Key Problems Identified:

- Delayed ticket resolution
- Data inconsistency
- Invalid status transitions
- Limited visibility for management

3. Project Objectives

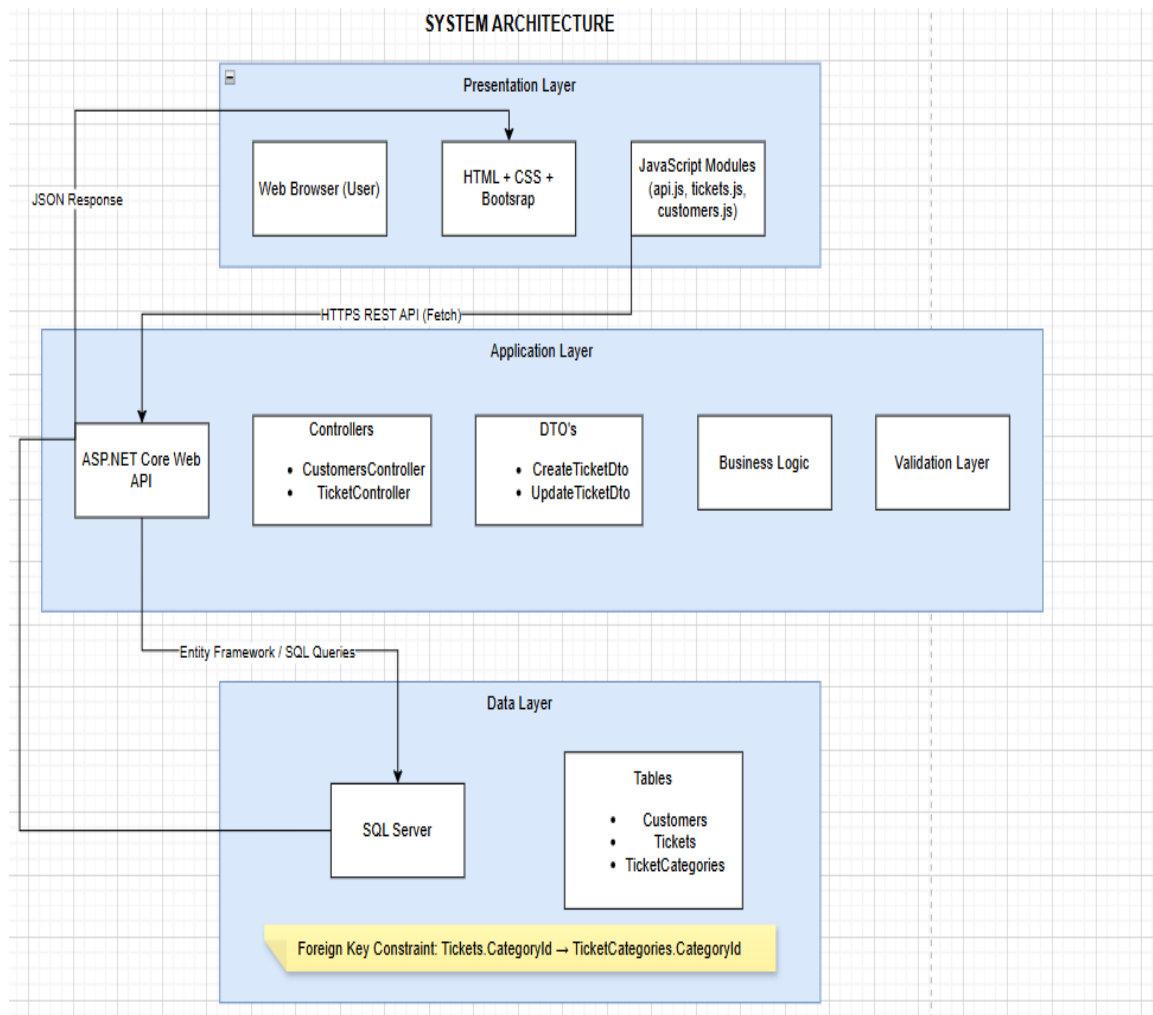
- Centralized ticket and customer management
- Enforced status workflow (Open → In Progress → Resolved)
- Dashboard analytics
- Secure and validated data entry
- Scalable architecture

4. System Architecture

Three-Tier Architecture:

1. Presentation Layer – HTML5, CSS3, Bootstrap, JavaScript
2. Application Layer – ASP.NET Core Web API
3. Data Layer – SQL Server Database

The frontend communicates with the backend via RESTful APIs over HTTPS. The backend processes business logic and interacts with the database through Entity Framework.



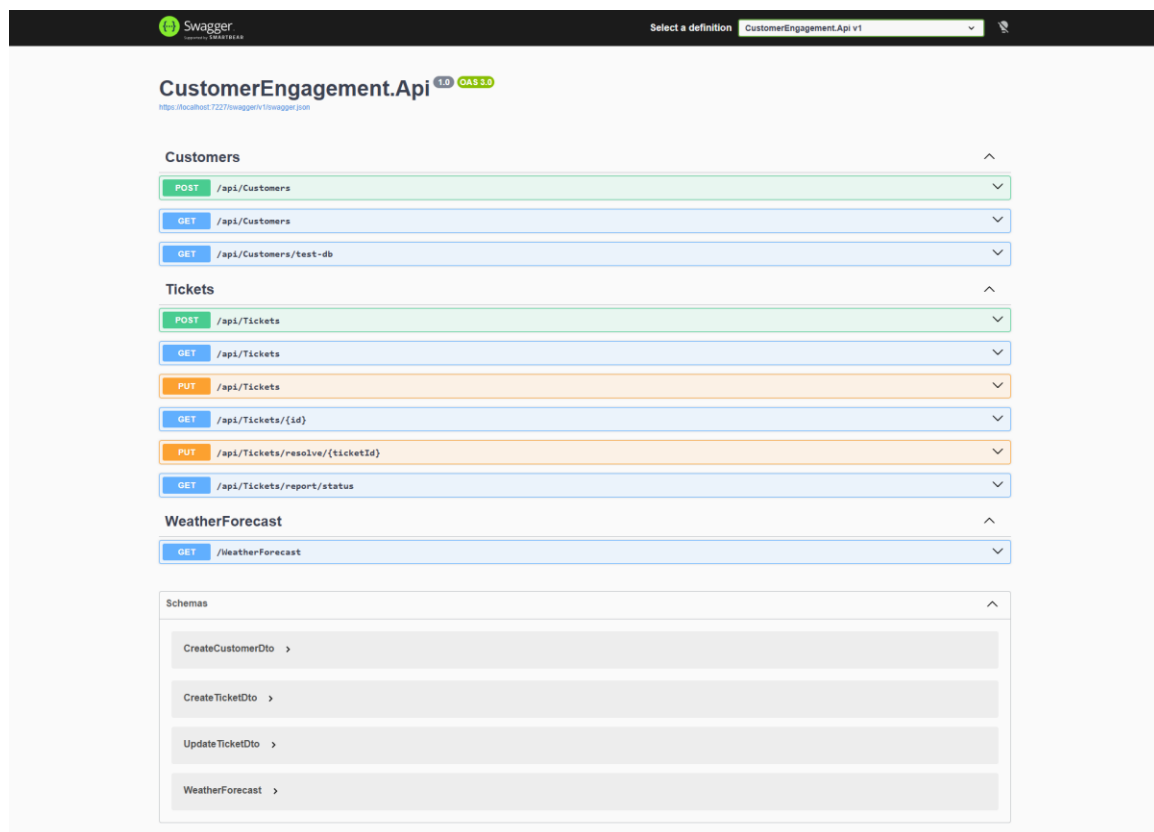
5. Technology Stack

Frontend: HTML5, CSS3, Bootstrap 5, Bootstrap Icons, JavaScript (ES6)

Backend: ASP.NET Core Web API (.NET 10)

Database: SQL Server

Tools: Swagger UI, Draw.io, VS Code, Visual Studio



6. Database Design & Schema

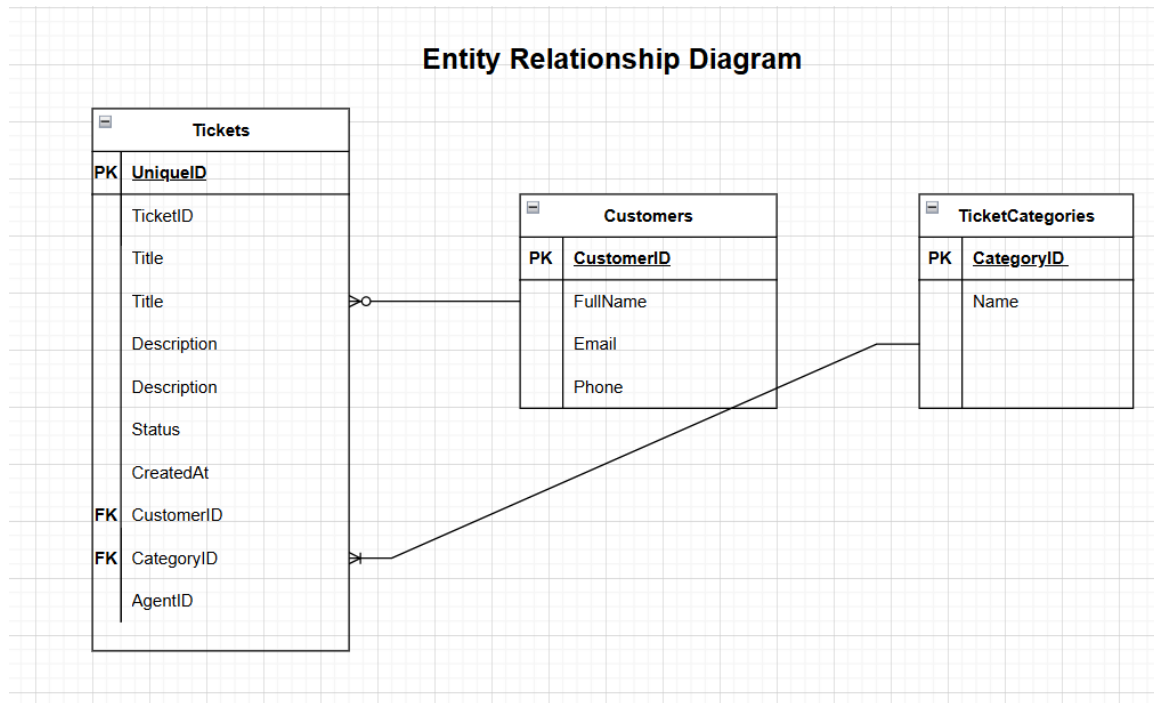
Core Tables:

Customers (CustomerId, FullName, Email, Phone)

Tickets (TicketId, Title, Description, Status, CategoryId, CustomerId, AgentId, CreatedAt)

TicketCategories (CategoryId, Name)

Foreign Key Constraints ensure Tickets reference valid CategoryIds. This prevents invalid category insertion and enforces referential integrity.



7. Backend Design & API Implementation

Implemented Endpoints:

POST /api/Customers

GET /api/Customers

POST /api/Tickets

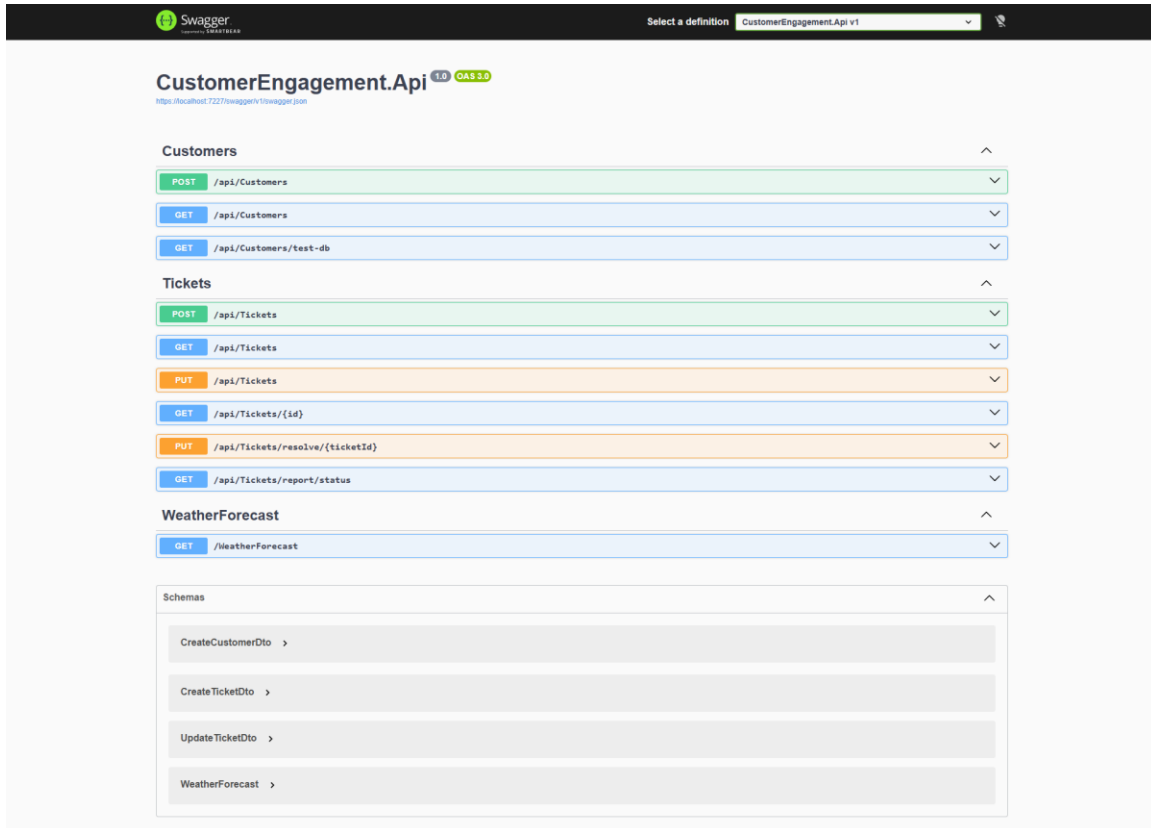
GET /api/Tickets

PUT /api/Tickets

PUT /api/Tickets/resolve/{ticketId}

GET /api/Tickets/report/status

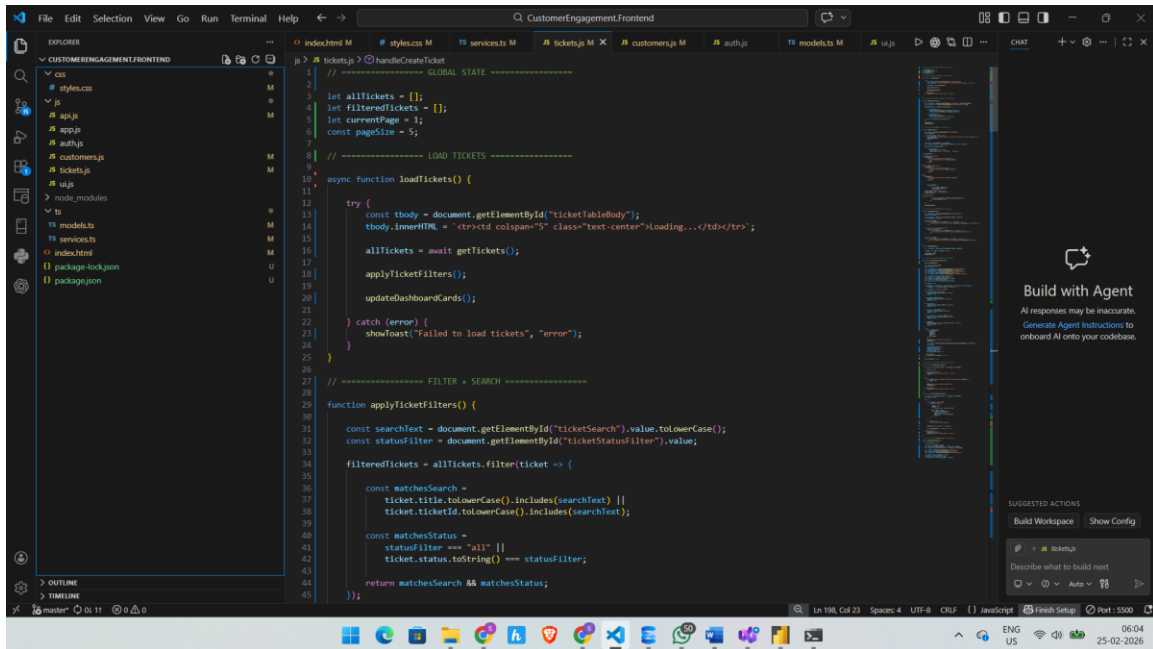
DTOs (CreateTicketDto, UpdateTicketDto) enforce structured updates.
Status transitions are validated server-side.



8. Frontend Architecture & UI Design

Modular JavaScript Files:

- api.js – Centralized API handler
- tickets.js – Ticket logic, pagination, filters
- customers.js – Customer operations
- ui.js – Toast & UI utilities
- app.js – Initialization logic



Features:

- Gradient dashboard cards
- Status-based action buttons
- Search & filtering
- Pagination
- Modal-based creation forms

9. Ticket Workflow & Business Rules

Status Lifecycle:

Open → In Progress → Resolved

Rules:

- Only Open tickets can move to In Progress
- Only In Progress tickets can be Resolved

- Resolved tickets cannot revert

Start →

Create Ticket (Status = Open)

↓

Decision: Agent Starts?

Yes → Status = In Progress

↓

Decision: Issue Resolved?

Yes → Status = Resolved

↓

End

The screenshot displays the 'Customer Engagement Dashboard' in a web browser. The dashboard includes four summary cards: 'Total Tickets' (10), 'Open Tickets' (4), 'Resolved Tickets' (3), and 'Total Customers' (60). Below these is a 'Tickets' tab with a search bar and a 'Create Ticket' button. A table lists tickets with columns for 'Ticket Id', 'Title', 'Status', 'Created At', and 'Action'. A dropdown menu is open over the 'Status' column, showing options: 'All Status', 'Open', 'In Progress', and 'Resolved'. The table shows tickets with statuses 'Open', 'Open', 'Resolved', and 'Resolved'. The 'Action' column contains buttons like 'Resolve', 'Start', and 'Resolved'.

Ticket Id	Title	Status	Created At	Action
17e527c3-b069-4009-ad07-0bf2d8ca5b1a	fghw	Open	24/02/2026, 21:39:48	Resolve
1785722e-0133-4e9b-9969-0eee2366bce5	jhhasbhb	Open	24/02/2026, 21:25:18	Start
eee09ce3-008d-47ef-b8fc-8728b3582f2c	Product inquiry	Open	24/02/2026, 21:18:28	Start
24f80a67-ba59-4d9c-a264-314e69c05996	Product inquiry	Resolved	24/02/2026, 21:18:28	Resolved
2fa8b1ff-5885-4107-a5d4-25986a4b06b7	Product inquiry	Resolved	24/02/2026, 21:18:28	Resolved

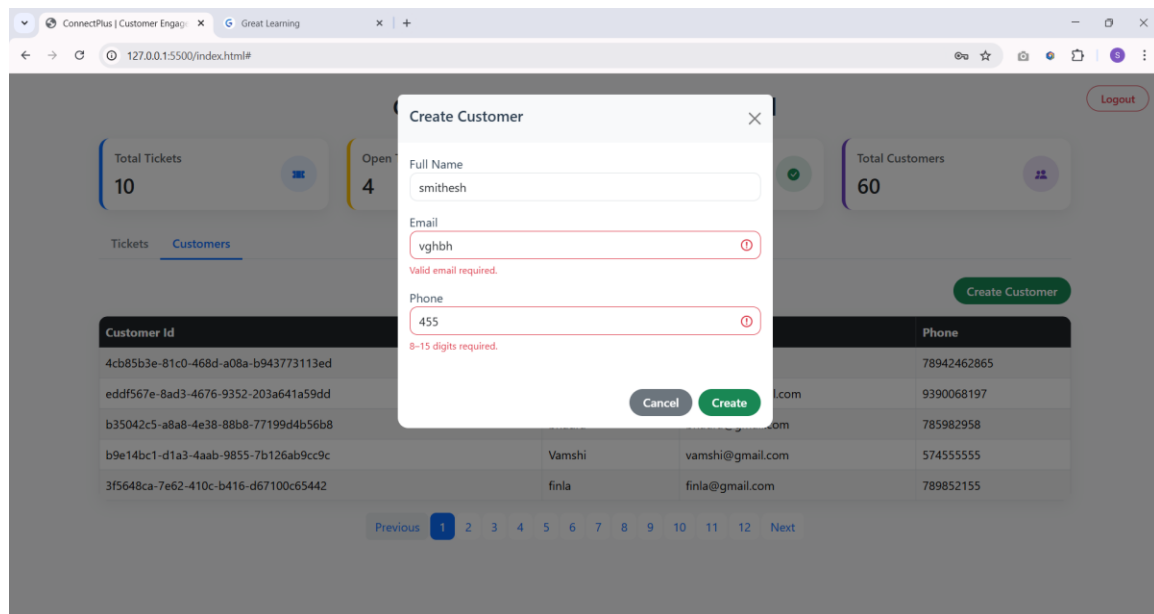
10. Validation & Error Handling Strategy

Frontend Validation:

- GUID format validation
- Minimum length checks
- Category dropdown to prevent FK errors

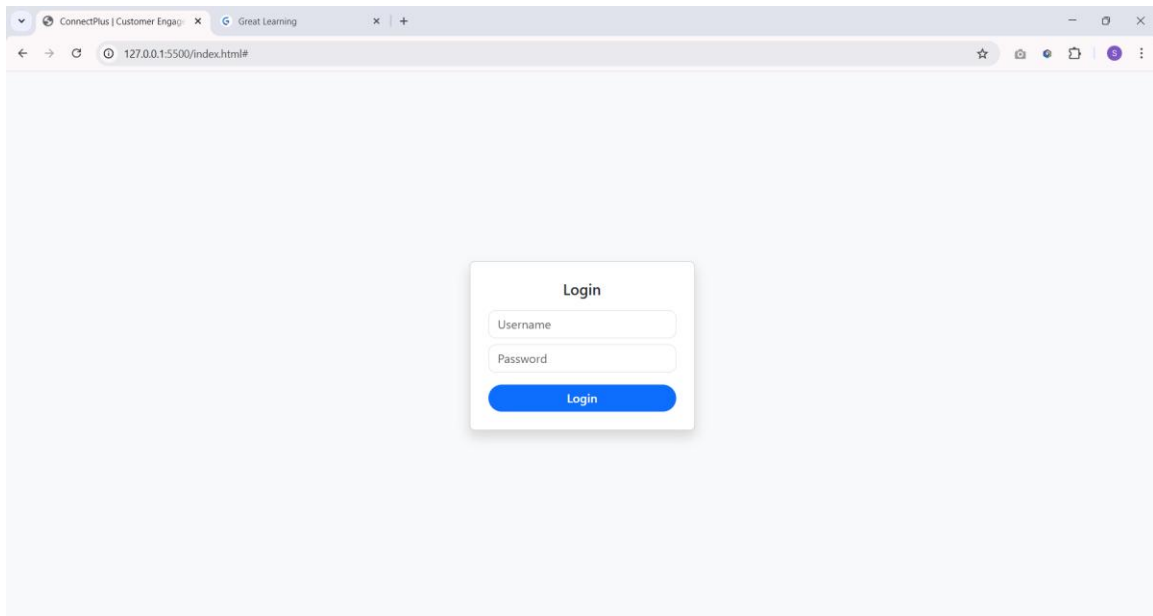
Backend Validation:

- DTO model validation
- SQL FK constraint enforcement
- HTTP status code management



11. Security & Authentication

- Login authentication module
- Controlled API access
- HTTPS-based communication
- Input validation to prevent malformed data



12. Analytics & Dashboard Logic

Dashboard Cards:

- Total Tickets
- Open Tickets
- In Progress Tickets
- Resolved Tickets
- Total Customers

Analytics are dynamically calculated from ticket dataset and rendered on UI.

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Customer Engagement Dashboard

Logout

Total Tickets10

Open Tickets4

Resolved Tickets3

Total Customers60

TicketsCustomers

Search tickets...All Status

Create Ticket

Ticket Id	Title	Status	Created At	Action
17e527c3-b069-4009-ad07-0bf2d8ca5b1a	fghwshghsg	In Progress	24/02/2026, 21:39:48	Resolve
1785722e-0133-4e9b-9969-0eee2366bce5	jhasbhb	Open	24/02/2026, 21:25:18	Start
eee09ce3-008d-47ef-b8fc-8728b3582f2c	Product inquiry	Open	24/02/2026, 21:18:28	Start
24f80a67-ba59-4d9c-a264-314e69c05996	Product inquiry	Resolved	24/02/2026, 21:18:28	Resolved
2fa8b1ff-5885-4107-a5d4-25986a4b06b7	Product inquiry	Resolved	24/02/2026, 21:18:28	Resolved

Previous12Next

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Customer Engagement Dashboard

Logout

Total Tickets10

Open Tickets4

Resolved Tickets3

Total Customers60

TicketsCustomers

Create Customer

Customer Id	Full Name	Email	Phone
4cb85b3e-81c0-468d-a08a-b943773113ed	hasi nbah	hasi@gmail.com	78942462865
eddf567e-8ad3-4676-9352-203a641a59dd	Smithb dgh	vchbjggb@gmail.com	9390068197
b35042c5-a8a8-4e38-88b8-77199d4b56b8	bhadra	bhadra@gmail.com	785982958
b9e14bc1-d1a3-4aab-9855-7b126ab9cc9c	Vamshi	vamshi@gmail.com	574555555
3f5648ca-7e62-410c-b416-d67100c65442	finla	finla@gmail.com	789852155

Previous123456789101112Next

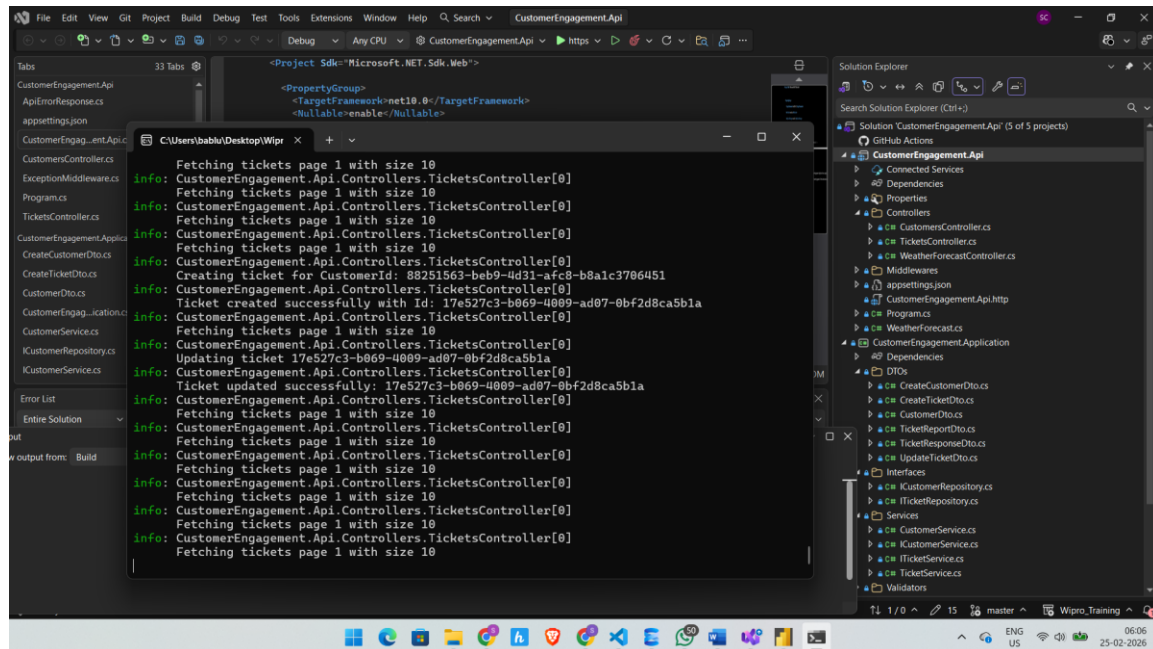
13. Deployment Strategy

Backend hosted locally via Kestrel server.

Frontend served using Live Server.

Database hosted on SQL Server instance.

Future deployment can use Azure App Service and Azure SQL.



14. Testing & Quality Assurance

- Swagger API Testing
- Manual UI Testing
- Validation Edge Case Testing
- Foreign Key Constraint Testing

15. Challenges Faced & Resolutions

- Foreign key constraint errors – resolved via category dropdown.
- Status update not reflecting – resolved by correct PUT payload structure.
- API 400 handling – improved centralized apiRequest error handling.

16. Future Enhancements

- Role-based access control
- Real-time notifications
- Advanced analytics charts
- CRM integration
- Cloud-native deployment

17. Power BI Dashboard Implementation

17.1 Data Connection

Power BI Desktop connects to SQL Server instance DESKTOP-HAGBT7R\SQLEXPRESS01 and imports the CustomerEngagementDB database tables.

17.2 KPI Measures (DAX)

Calculated metrics implemented:

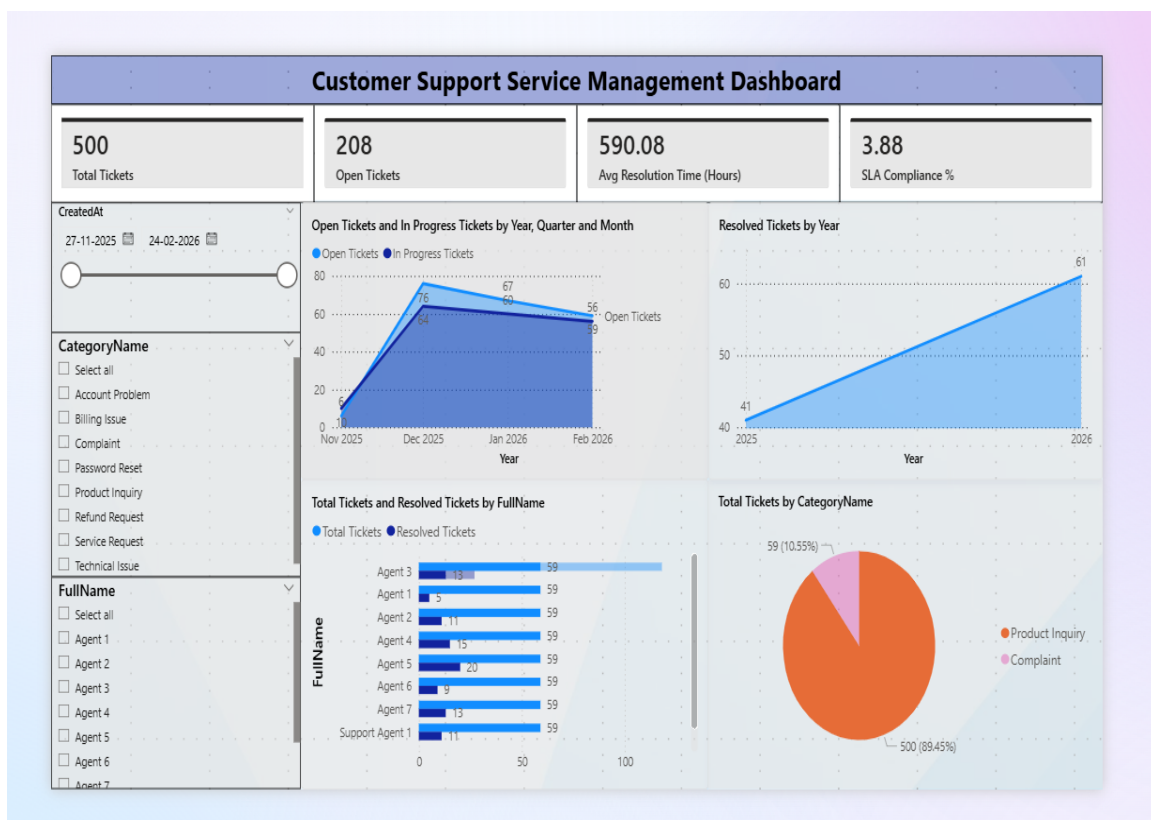
- Total Tickets
- Open Tickets (Backlog)

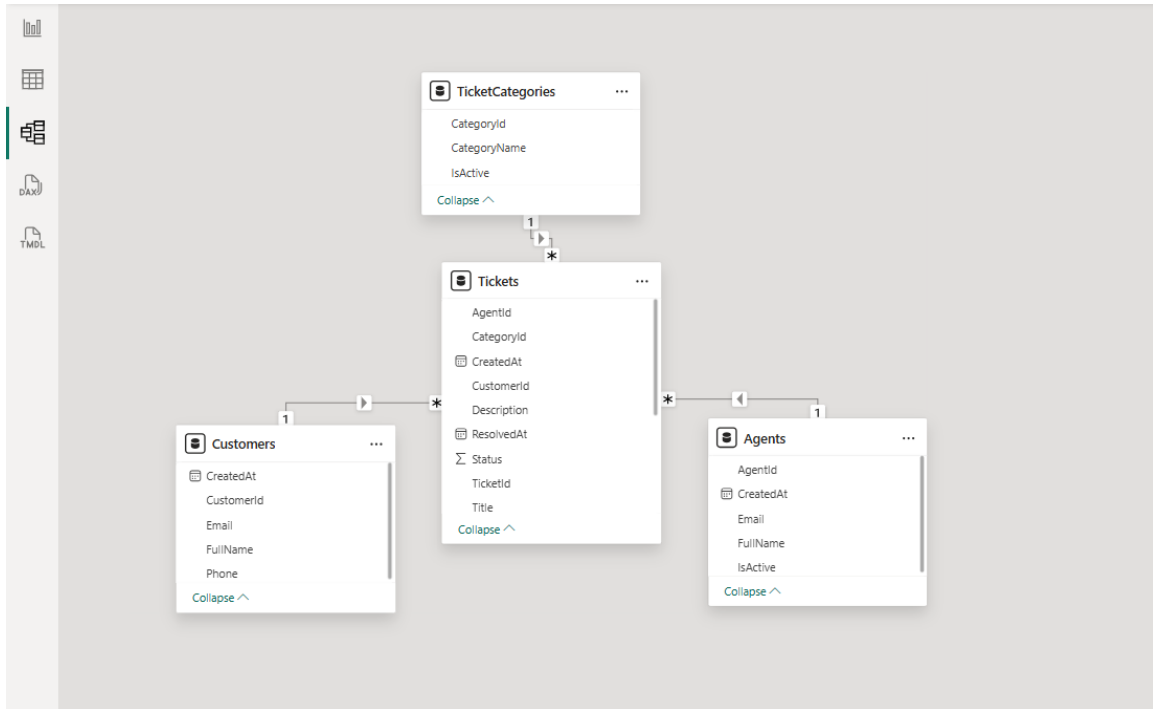
- In Progress Tickets
- Resolved Tickets
- Average Resolution Time
- SLA Compliance Percentage

17.3 Dashboard Visuals

Executive dashboard includes:

- KPI Cards — high-level performance indicators
- Backlog Trend Line Chart — Open + In Progress tickets over time
- Resolution Trend — resolved tickets progression
- Agent Performance Bar Chart — workload comparison
- Category Distribution Pie Chart — issue type analysis





17.4 Interactive Filters (Slicers)

Dashboard includes slicers for dynamic analysis:

- Date Range (CreatedAt)
- Agent (FullName)
- Category (CategoryName)

These filters allow management to analyze performance across dimensions.

17.5 Layout & Design

A professional executive layout was applied:

- Top Row: KPI Cards
- Left Panel: Slicers
- Middle: Trend Charts
- Bottom: Performance Visuals

Minimal background and Segoe UI font used for clarity.

18. Insights Supported by Dashboard

The dashboard enables management to monitor service efficiency, backlog trends, and agent productivity, supporting data-driven decision making and service improvement planning.

19. Conclusion

The implemented analytics solution provides a comprehensive management view of customer support operations. It integrates structured operational data with interactive visual analytics to support monitoring, evaluation, and strategic decision-making. The system successfully delivers a secure, scalable, and enterprise-grade Customer Engagement platform aligned with business modernization goals.